

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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MODEL PRACTICAL EXAMINATION – SET 5

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Flight Reservation System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as one way, round trip, From, To, Departure, Travelers & Class, Fare Type, International and Domestic Flights, etc.
2. Create a simple cloud software application for Property Buying & Rental process (in Chennai city) using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Buy, Rent, Commercial, Rental Agreement, Area, Place, Range, Property Value, etc.
3. Create a Snapshot of a VM and test it by loading the previous version/cloned VM.
4. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of your present Windows environment.

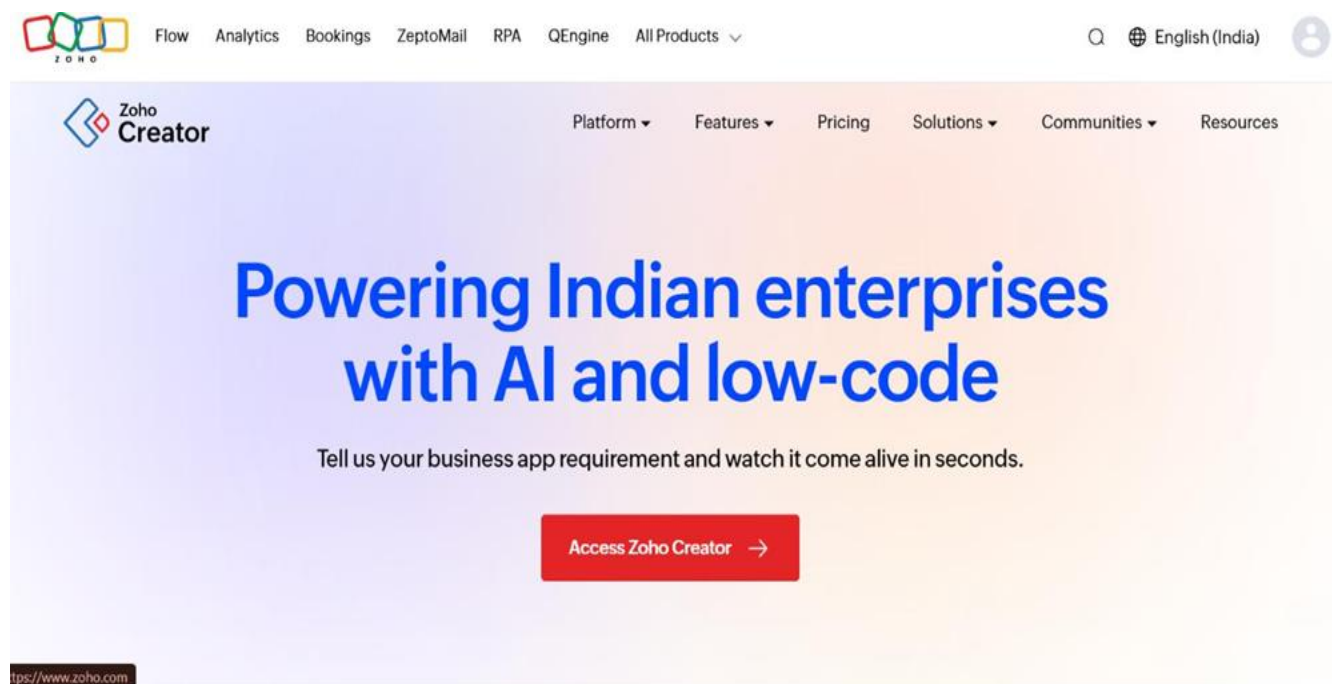
EXPERIMENT 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR FLIGHT BOOKING RESERVATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

AIM:

To Create a simple cloud software application for Flight Booking Reservation System using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

STEP1: Log into zoho.com



STEP 2: Create new application from scratch



Create from scratch

Application Name *

☐ Enable Environment [Learn More](#)

Create Application

STEP 3: Create new application and create form

≡

Flight Reservation System

+

 Design

 Workflow

 Settings

 Upgrade



Flight Reservation System



Start building your application by creating a form

Create New Form

How would you like to create your form?

On my own

+

From scratch

⇅

Import with data

Zia

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50+ View More

Using an Integrated Datasource

db

cloud

link

red

color

...

STEP 4: Create Bookings Form

Flight Reservation System

Bookings

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Booking ID

Passenger

Flight

Booking Date

Seat Number

Meal Preference

Extra Baggage

Field Properties

Field name

Booking ID

Field link name

booking_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

STEP 5: Create Flights Form

Flight Reservation System

Bookings

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Booking ID

Passenger

Flight

Booking Date

Seat Number

Meal Preference

Extra Baggage

Field Properties

Field name

Booking ID

Field link name

booking_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

STEP 6: Create Passengers Form

Flight Reservation System

Passengers

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Passenger ID

First Name

Last Name

Gender

Date of Birth

Email

Mobile Number

Aadhaar Number

Field Properties

Field name

Passenger ID

Field link name

passenger_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

Chats

Contacts

Here is your Smart Chat (Ctrl+Space)

Trial expires in 15 days | Upgrade

Get started faster!

Help

STEP 7: Create Payments Form

Flight Reservation System

Payments

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Booking

Payment Date

Payment Method

Transaction ID

Amount

Status

Field Properties

Field name

Booking

Field link name

booking

View Field References

Display Fields

Flight Reservation System → Bookings

Booking ID

Display type

Multi Select

Here is your Smart Chat (Ctrl+Space)

Trial expires in 15 days | Upgrade

Get started faster!

Help

STEP 8: Create Feedback Form

Flight Reservation System

Feedbacks

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Passenger

Flight

Rating

Review

Would Recommend

Field Properties

Field name

Passenger

Field link name

passenger

View Field References

Display Fields

Flight Reservation System → Passengers

Passenger ID

Display type

Multi Select

Here is your Smart Chat (Ctrl+Space)

Trial expires in 15 days | Upgrade

Get started faster!

Help

9: Create Search History Form

The screenshot shows the 'Search History' configuration page in the Flight Reservation System. The interface is divided into three main sections: 'Basic Fields', 'Field Properties', and a bottom status bar.

Basic Fields: This section contains a grid of field icons and a list of selected fields. The icons include Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date. The selected fields are:

- Trip Type (highlighted in blue)
- Flight Category
- From City
- Destination
- Departure Date (highlighted in blue)
- Return Date
- Travellers

Field Properties: This section shows the configuration for the selected 'Trip Type' field.

- Field name:** Trip Type
- Field link name:** trip_type
- View Field References:** A link to view references.
- Choices:** A section for defining choices for the field. It includes a note: "Renaming the choices will update the values in the existing records, if any." Below this are two radio button options: "One Way" and "Round Trip", each with plus, minus, and edit icons.

Status Bar: At the bottom, there is a chat icon, a message "Here is your Smart Chat (Ctrl+Space)", a trial expiration notice "Trial expires in 15 days | Upgrade", and a "Get started faster! Help" link.

STEP 10: Add sample data to all the created forms.

The screenshot shows the 'Feedbacks' form in the Flight Reservation System. The interface includes a sidebar menu, a header bar, and a main form area.

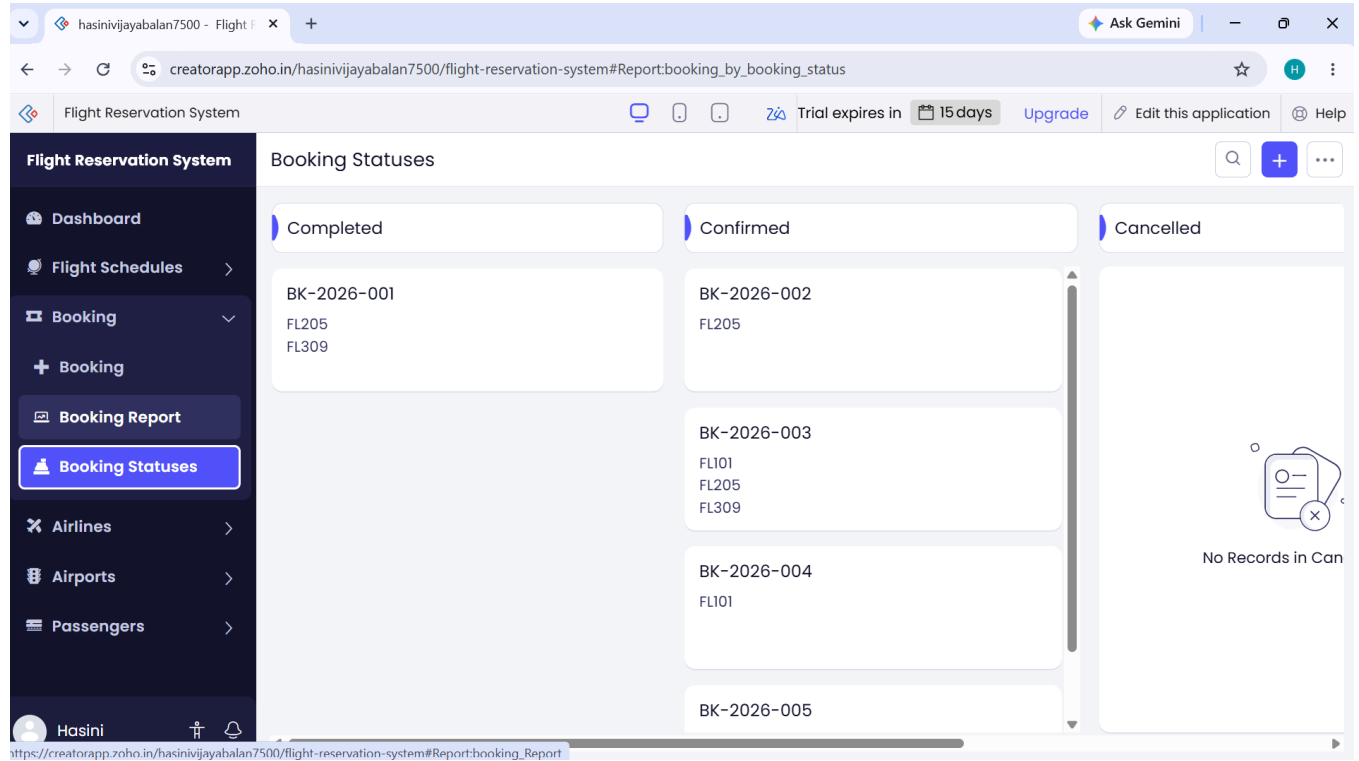
Sidebar Menu: The menu items are: Dashboard, Bookings, Payments, Feedbacks (selected), Feedbacks Report, Flights, Passengers, and Search History.

Header Bar: The header bar displays the system name "Flight Reservation System", navigation icons, a trial expiration notice "Trial expires in 14 days | Upgrade", and links for "Edit this application" and "Help".

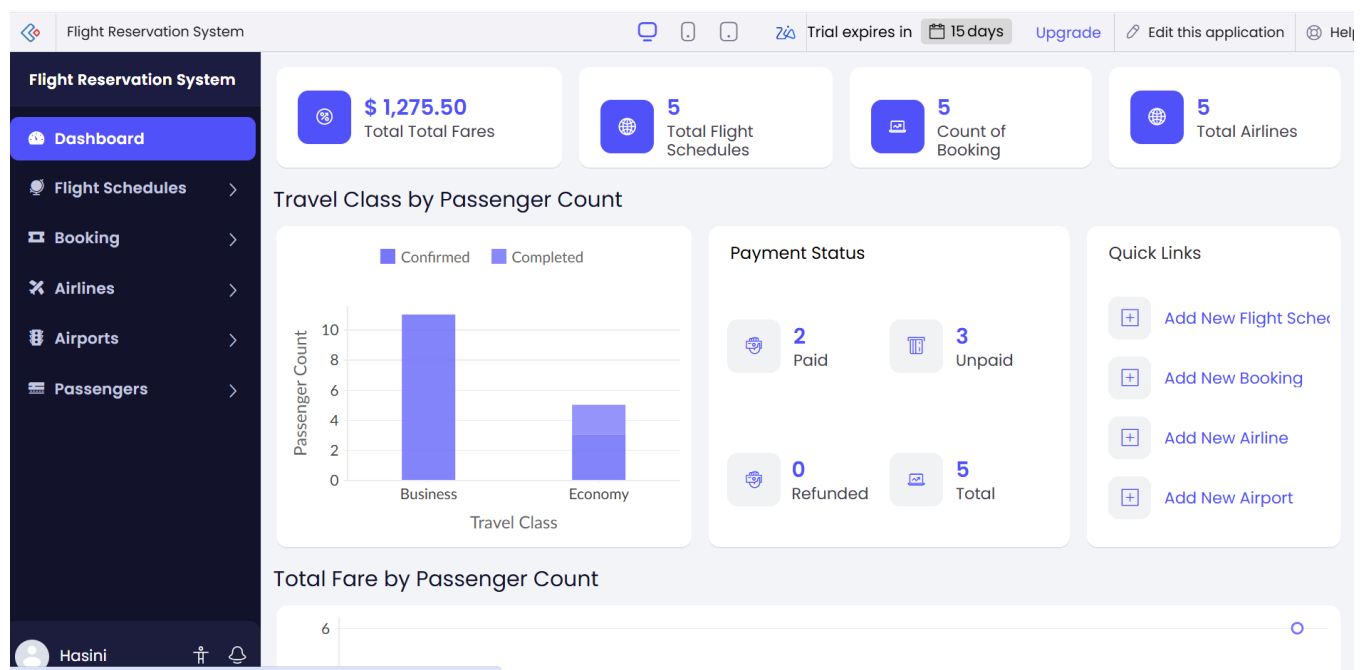
Feedbacks Form: The form contains the following fields:

- Passenger:** A text input field containing "4 X".
- Flight:** A text input field containing "5 X".
- Rating:** A text input field containing "4".
- Review:** A text area containing "GOOD".
- Would Recommend:** A section with three radio button options: "No", "Yes" (selected), and "Other".

STEP 11: Create reports to display the data entered in the respective forms.



STEP 12: Create and display the Dashboard with important information and charts from the collected data.

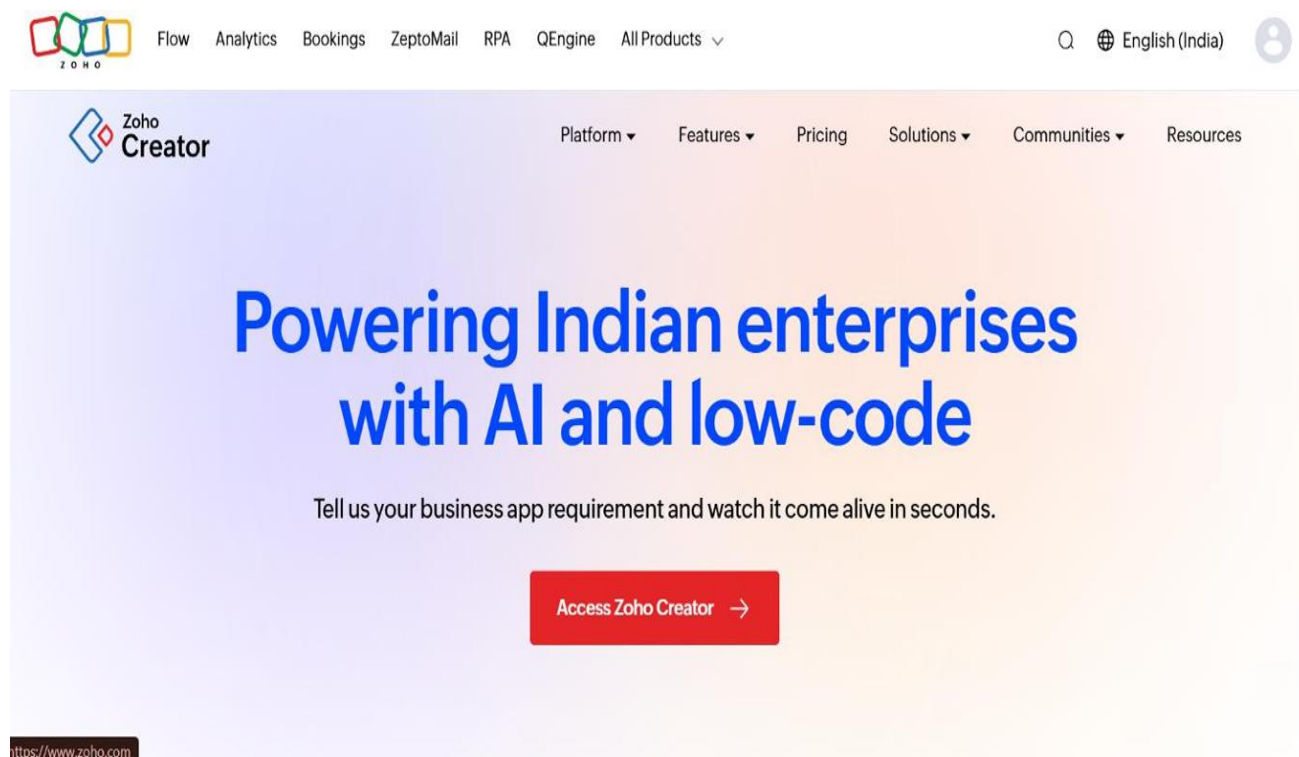


EXPERIMENT 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PROPERTY BUYING & RENTAL PROCESS IN CHENNAI CITY USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SAAS.

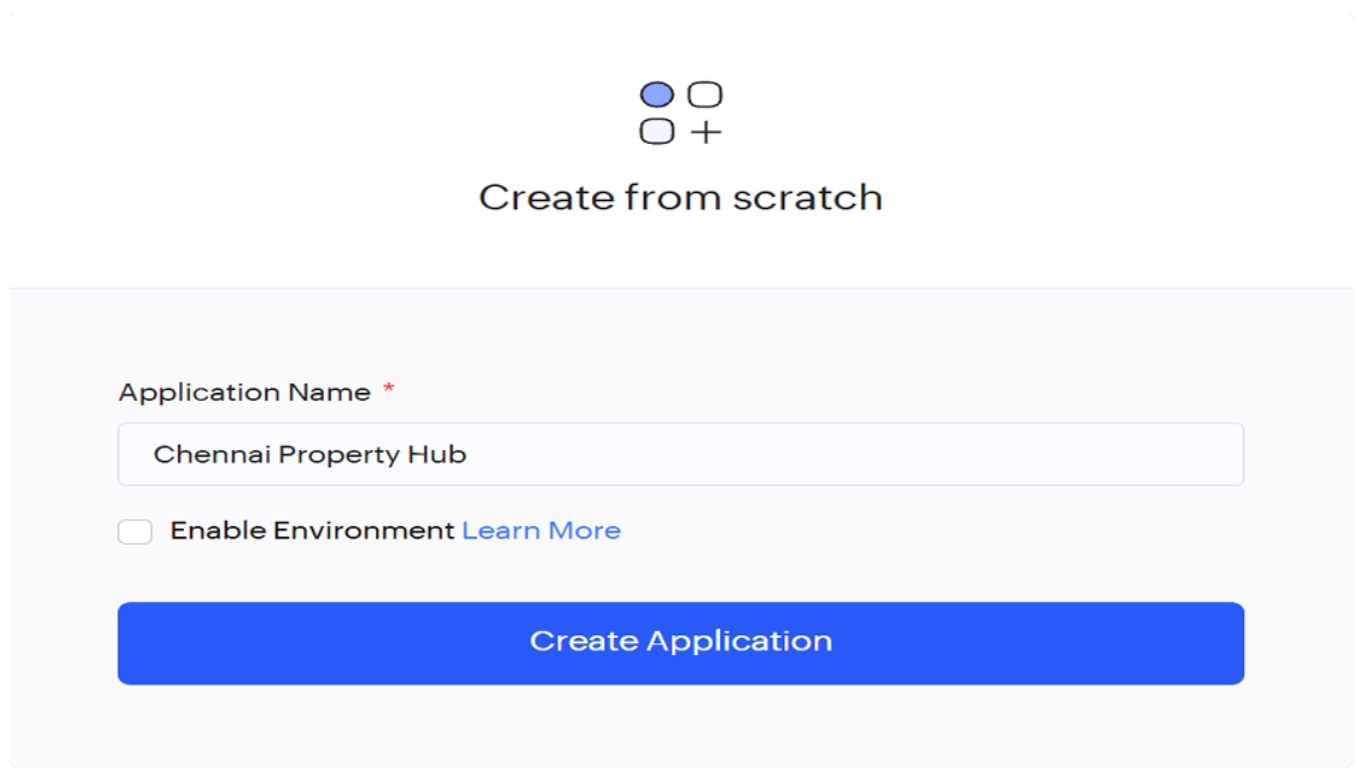
Aim: To create a simple cloud software application for Property Buying and Rental Process in Chennai City using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

STEP 1: Log into zoho.com.

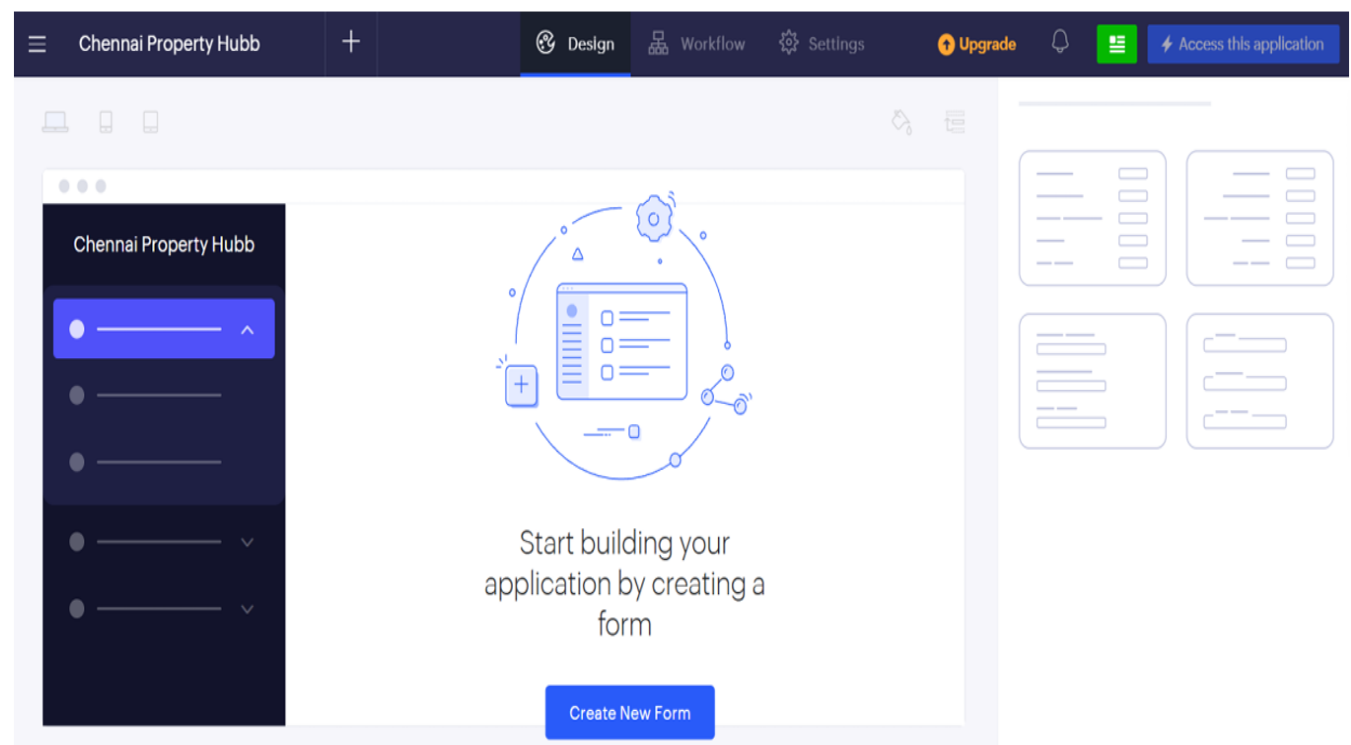


STEP 2: Create a new application from scratch in Zoho Creator.



The screenshot shows the 'Create from scratch' interface in Zoho Creator. At the top, there is a logo consisting of four squares (two blue, two white) and a plus sign. Below the logo, the text 'Create from scratch' is displayed. The main form area has a label 'Application Name' with a red asterisk. Below this is a text input field containing 'Chennai Property Hub'. Under the input field, there is a checkbox labeled 'Enable Environment' followed by a blue link 'Learn More'. At the bottom of the form is a large blue button labeled 'Create Application'.

STEP 3: Create a new application and create forms from scratch.



The screenshot shows the Zoho Creator application design interface. The top navigation bar includes a hamburger menu, the application name 'Chennai Property Hubb', a plus icon, and tabs for 'Design', 'Workflow', and 'Settings'. There are also links for 'Upgrade', a notification bell, a list icon, and 'Access this application'. The main workspace is divided into three sections. On the left is a dark sidebar with the title 'Chennai Property Hubb' and a list of form thumbnails. The center section features a large circular diagram with a central form icon and the text 'Start building your application by creating a form'. Below this text is a blue button labeled 'Create New Form'. On the right side, there are four form thumbnails arranged in a 2x2 grid.

How would you like to create your form?

On my own

+

From scratch

Import with data

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 + View More

Using an Integrated Datasource

ab

...

STEP 4: Create Property Form.

Chennai Property Hub

Properties

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

Date

Property ID

Property Name

Property Type

Purpose

Category

123 Area

Location

Chennai Zone

Field Properties

Field name

Property ID

Field link name

property_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days

Upgrade

Get started faster!

Help

STEP 5: Create Customers Form.

Chennai Property Hub

Customers

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Customer ID

Customer Name

Gender

Mobile Number

Email

Address

Preferred Property Type

Status

Field Properties

Field name

Customer ID

Field link name

customer_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

STEP 6: Create Property Search Form.

Chennai Property Hub

Property Searches

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

123

Number

15

Date

Buy or Rent

Property Category

Preferred Location

Budget Range

Bedrooms

Furnishing

Customer

Field Properties

Field name

Buy or Rent

Field link name

buy_or_rent

View Field References

Choices

Advanced

Renaming the choices will update the values in the existing records, if any.

Buy

Rent

STEP 7: Create Rental Agreement Form.

Chennai Property Hub
Rental Agreements

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Customer

Property

Agreement Start Date

Agreement End Date

Monthly Rent

Security Deposit

Agreement Status

Field Properties

Field name
Customer

Field link name
customer

[View Field References](#)

Display Fields
Chennai Property Hub → Customers
Customer ID

Display type
Multi Select

Chat

Feedback

Here Is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

Get started faster!

Help

STEP 8: Create Property Bookings forms.

Chennai Property Hub
Property Bookings

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Customer

Property

Buy or Rent

Booking Date

Advance Amount

Total Property Value

Booking Status

Field Properties

Field name
Customer

Field link name
customer

[View Field References](#)

Display Fields
Chennai Property Hub → Customers
Customer ID

Display type
Multi Select

Chat

Feedback

Here Is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

Get started faster!

Help

STEP 9: Create Feedback Form

The screenshot displays the 'Chennai Property Hub' interface for configuring a 'Property Bookings' form. The top navigation bar includes a 'Done' button. The main area is divided into three sections: 'Basic Fields', 'Field Properties', and a bottom status bar.

Basic Fields: This section contains a grid of field icons for selection. The icons include: Name, Email, Address, Phone, Single Line, Multi Line, Number, and Date.

Field Properties: This section shows the configuration for the selected 'Customer' field. It includes:

- Field name:** Customer
- Field link name:** customer
- View Field References:** A link to view references.
- Display Fields:** A section showing the field's placement in the form, with a dropdown menu set to 'Customer ID'.
- Display type:** A dropdown menu set to 'Multi Select'.

Form Fields: The central area shows the form fields being configured:

- Customer:** A search field with a magnifying glass icon.
- Property:** A search field with a magnifying glass icon.
- Buy or Rent:** A radio button group with 'Buy' and 'Rent' options.
- Booking Date:** A date picker field.
- Advance Amount:** A text input field with a dollar sign icon.
- Total Property Value:** A text input field with a dollar sign icon.
- Booking Status:** A dropdown menu.

The bottom status bar includes a chat icon, a 'Here is your Smart Chat (Ctrl+Space)' message, a 'Trial expires in 14 days' notification, an 'Upgrade' button, a 'Get started faster!' button, and a 'Help' button.

STEP 10: Add sample data to all the created forms.

The screenshot displays the 'Chennai Property Hub' interface for the 'Property Bookings' form. The left sidebar shows the navigation menu with 'Property Bookings' selected. The main area shows the form with sample data entered.

Form Fields:

- Customer:** 1 x
- Property:** 4 x
- Buy or Rent:** Buy (radio button), Rent (radio button, selected), Other (radio button)
- Booking Date:** 28-Jul-2026
- Advance Amount:** 20000
- Total Property Value:** 100000
- Booking Status:** Confirmed (dropdown menu)

At the bottom of the form, there are two buttons: 'Submit' and 'Reset'.

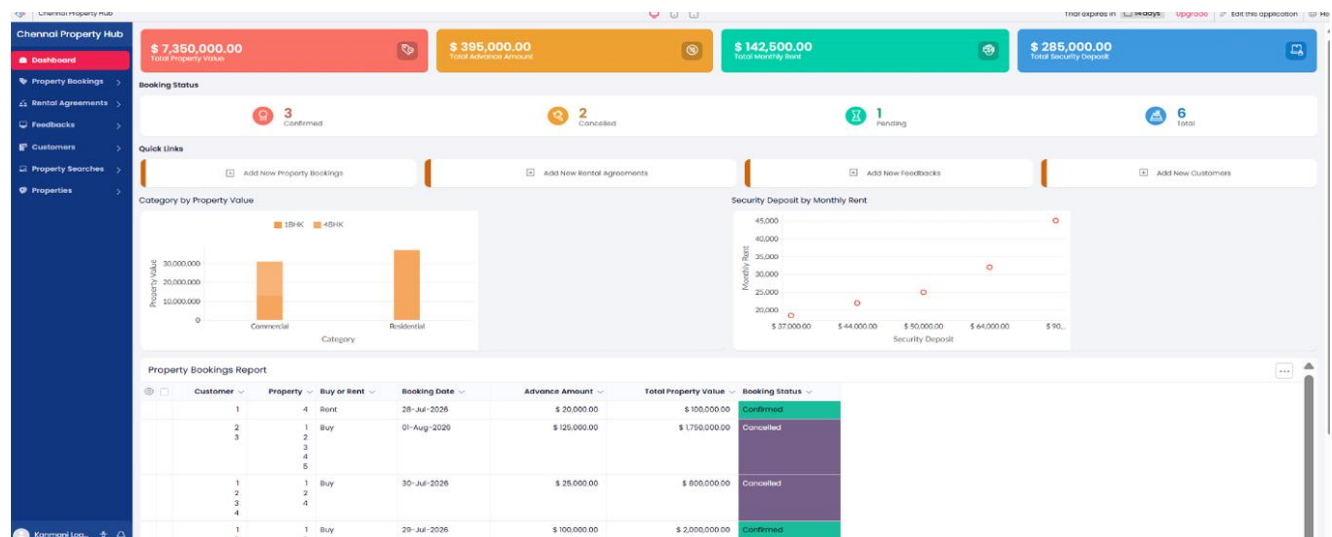
STEP 11: Create reports to display the data entered in the respective forms.

The screenshot shows the 'Property Bookings' form in the Chennai Property Hub application. The form includes the following fields and options:

- Customer:** A text input field containing '1'.
- Property:** A text input field containing '4'.
- Buy or Rent:** Radio button options for 'Buy', 'Rent' (selected), and 'Other'.
- Booking Date:** A date input field containing '28-Jul-2026'.
- Advance Amount:** A text input field containing '20000'.
- Total Property Value:** A text input field containing '100000'.
- Booking Status:** A dropdown menu showing 'Confirmed'.

At the bottom of the form are two buttons: 'Submit' (in red) and 'Reset'.

STEP 12: Create and display the Dashboard with important property details, buying and rental information, and charts from the collected data.



EXPNO 3: CREATE A SNAPSHOT OF A VM AND TEST IT BY LOADING THE PREVIOUS VERSION/CLONED VM

AIM:

To create a snapshot of a vm and test it by loading the previous version/cloned vm
PROCEDURE:

STEP 1: GOTO VMWARE WORKSTATION.

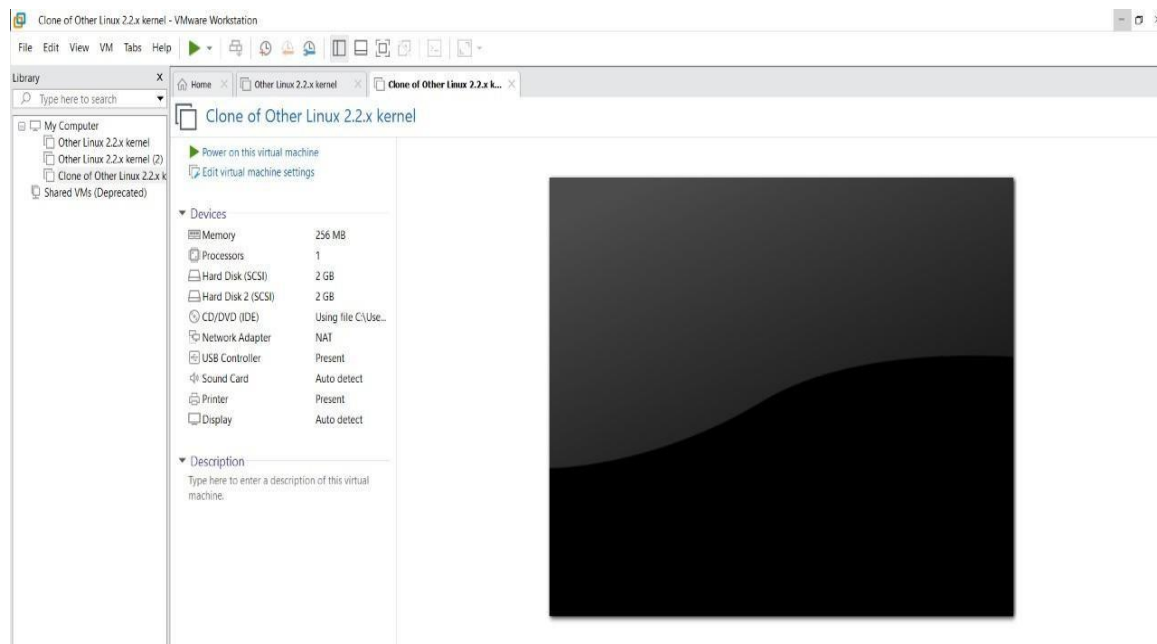
STEP 2: CREATE FILES ON DESKTOP.

STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.

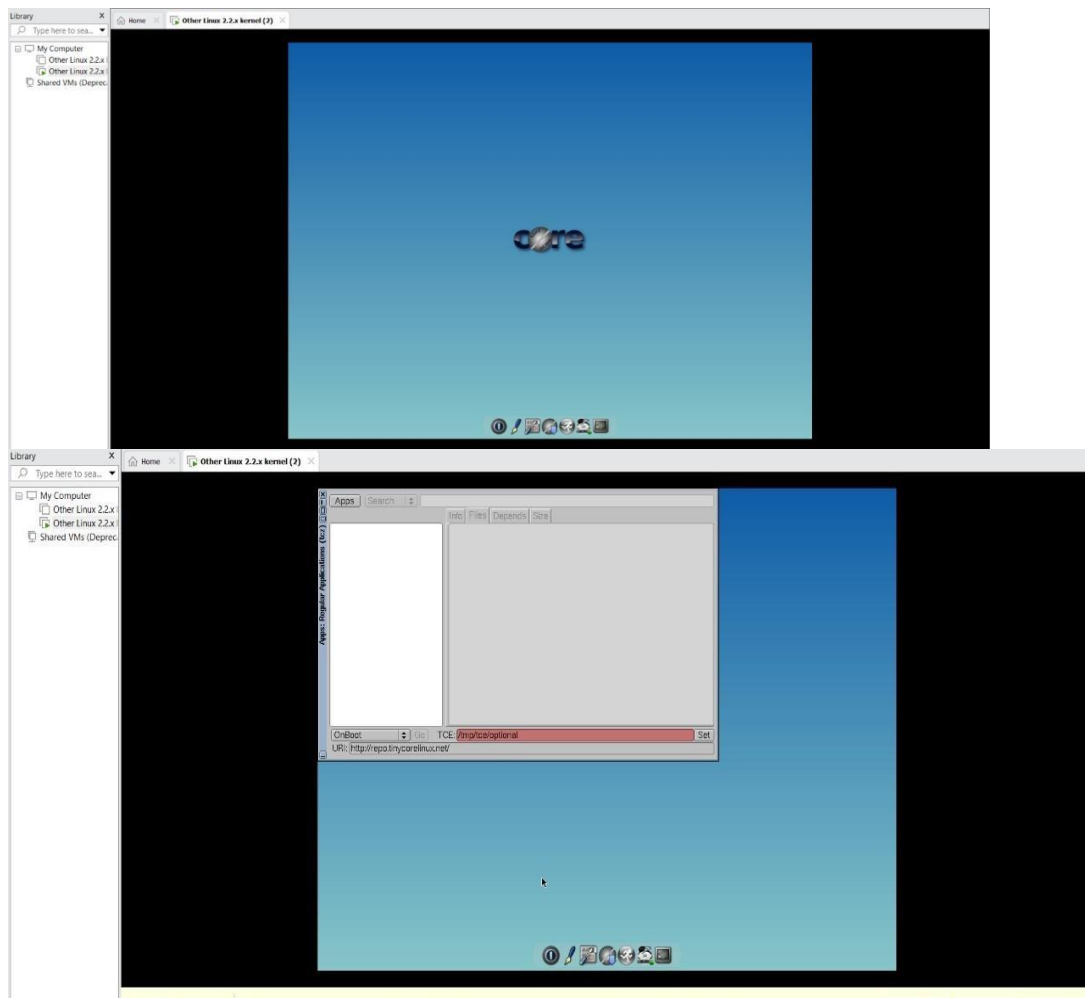
STEP 4: SNAPSHOT IS BEING DONE

IMPLEMENTATION:

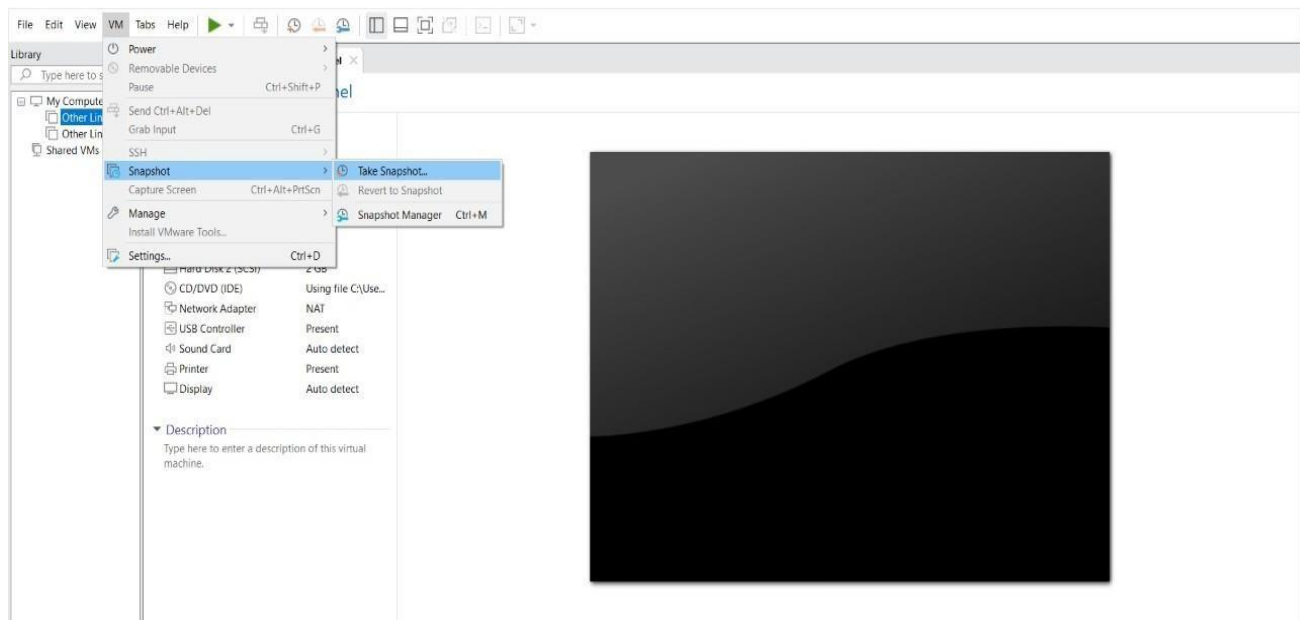
STEP 1: GOTO VMWARE WORKSTATION

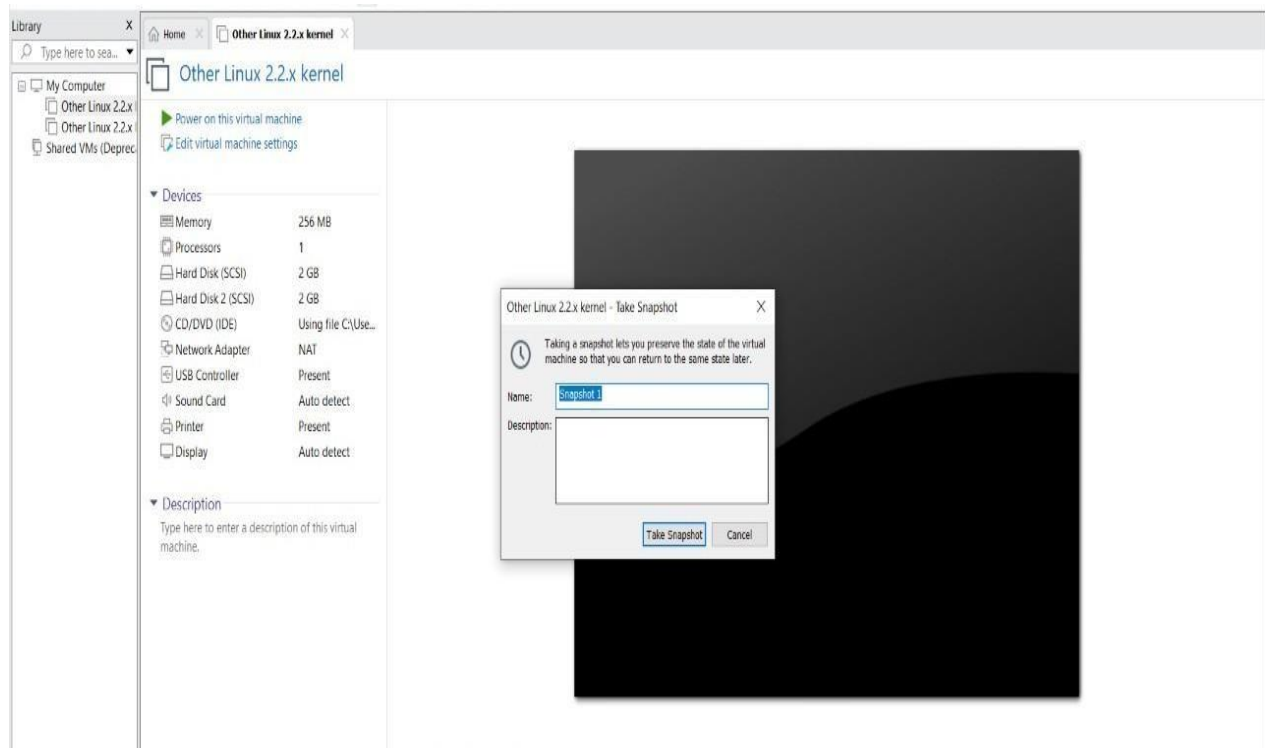


STEP 2: CREATE FILES ON DESKTOP

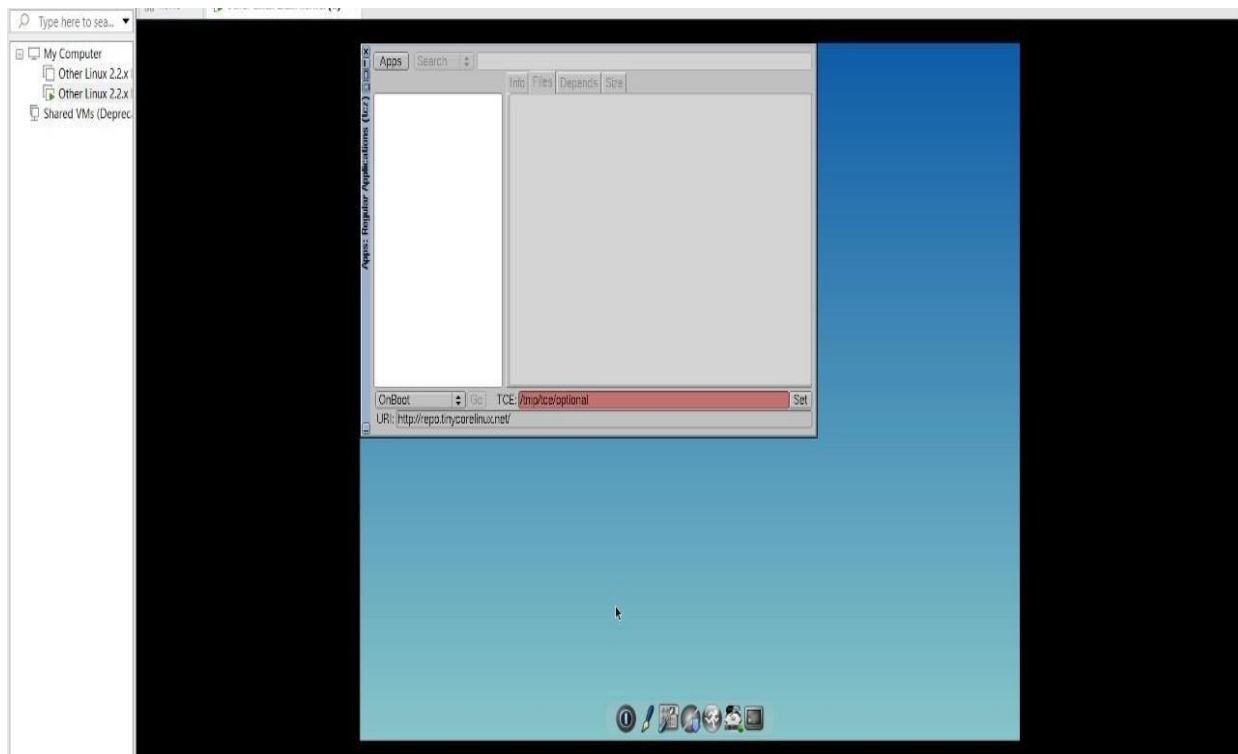


STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.





STEP 4: SNAPSHOT IS BEING DONE



EXPNO 4: CREATE A VIRTUAL MACHINE WITH 1 CPU, 2GB RAM AND 15GB STORAGE DISK USING A TYPE 2 VIRTUALIZATION SOFTWARE.

AIM:

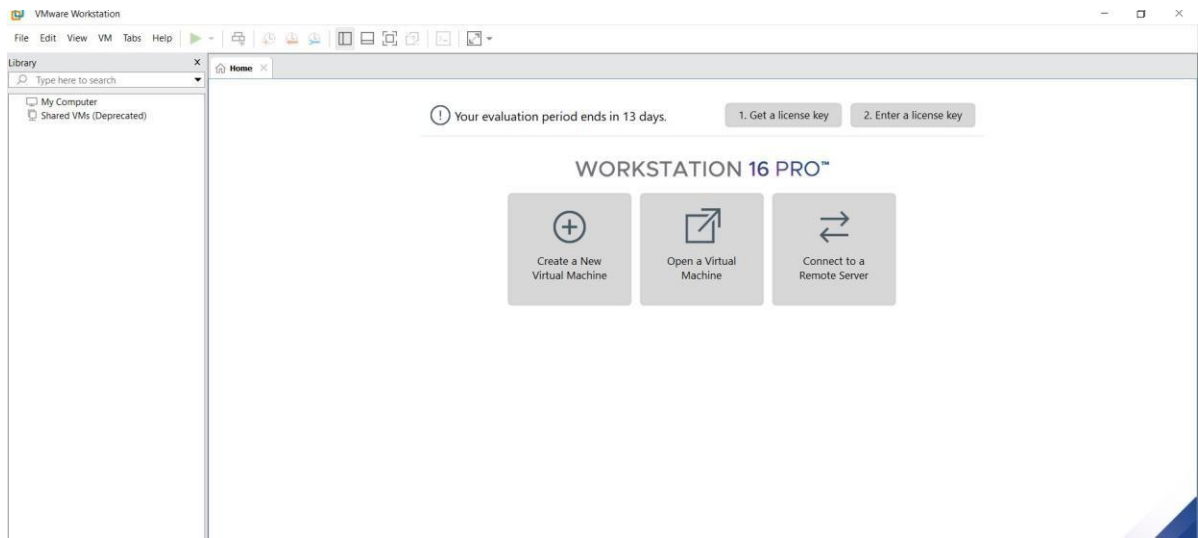
To create a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

PROCEDURE:

- STEP 1:Download VMware workstation and installed as type 2 hypervisor.
- STEP2:Download ubuntu or tiny OS as iso image file.
- STEP 3: In VMware workstation->create new VM.
- STEP 4: Do the basic configuration settings.
- STEP 5: Created tiny OS virtual machine.
- STEP 6: Launch the VM.

IMPLEMENTATION:

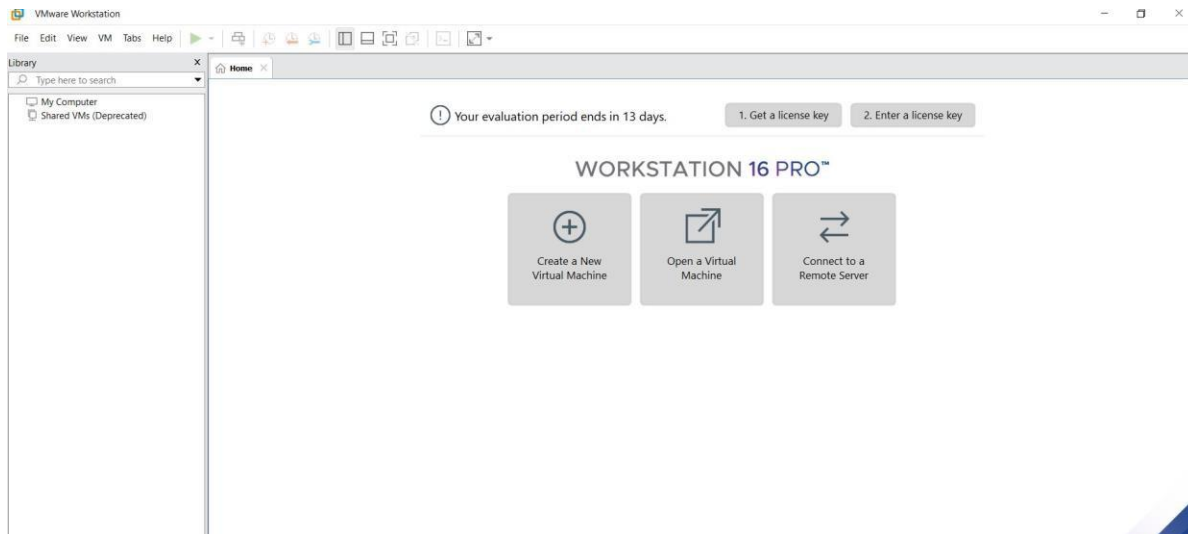
STEP 1:DOWNLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2 HYPERVISOR



STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

Index of /11.x/x86/release/			
..			
distribution_files/	09-Feb-2020 11:50	-	
src/	03-Dec-2019 11:14	-	
Core-11.1.iso	01-Apr-2020 07:49	14757888	
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48	
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639	
Core-current.iso	01-Apr-2020 07:49	14757888	
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656	
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52	
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358	
CorePlus-current.iso	01-Apr-2020 07:50	216006656	
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944	
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52	
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301	
TinyCore-current.iso	01-Apr-2020 07:50	19922944	

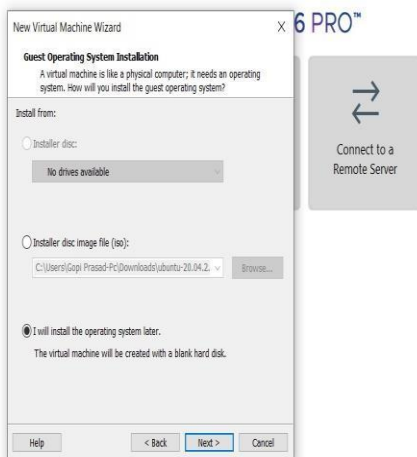
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



! Your evaluation period ends in 12 days. 1. Get a license key 2. Enter a license key



New Virtual Machine Wizard

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help < Back Next > Cancel

6 PRO™



Connect to a
Remote Server

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu Vmware

Location:

C:\Users\Gopi Prasad-PC\Documents\Virtual Machines\Ubuntu

Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

6 PRO™



Connect to a
Remote Server

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name: Ubuntu Vmware

Location: C:\Users\Gopi Prasad-Pc\Documents\Virtual Machines\U...

Version: Workstation 16.x

Operating System: Ubuntu 64-bit

Hard Disk: 20 GB

Memory: 2048 MB

Network Adapter: NAT

Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...

Customize Hardware...

< Back

Finish

Cancel



ⓘ Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

ⓘ Your evaluation period ends in 12 days.

1. Get a license key

2. Enter a license key

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file

☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

Connect to a Remote Server

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

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Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

Connect to a Remote Server

New Virtual Machine Wizard

Ready to Create Virtual Machine

Click Finish to create the virtual machine. Then you can install Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name: Ubuntu Vmware

Location: C:\Users\Gopi Prasad-Pc\Documents\Virtual Machines\U...

Version: Workstation 16.x

Operating System: Ubuntu 64-bit

Hard Disk: 20 GB

Memory: 2048 MB

Network Adapter: NAT

Other Devices: 2 CPU cores, CD/DVD, USB Controller, Printer, Sound C...

Customize Hardware...

< Back

Finish

Cancel

Connect to a Remote Server

STEP 5: CREATED TINYOS VIRTUAL MACHINE

